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- 1 Siderowf A, Concha-Marambio L, Lafontant DE, et al. Assessment of heterogeneity among participants in the Parkinson's Progression Markers Initiative cohort using α -synuclein seed amplification: a cross-sectional study. *Lancet Neurol* 2023; **22**: 407–17.
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- 4 Chahine LM, Merchant K, Siderowf A, et al. Proposal for a biologic staging system of Parkinson's disease. *J Parkinsons Dis* 2023; **13**: 297–309.
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We applaud Andrew Siderowf and colleagues for presenting the first rigorous and comprehensive evaluation of an α -synuclein seed amplification assay (SAA) in CSF.¹ Rigour was optimised by use of CSF from individuals with Parkinson's disease enrolled in the Parkinson's Progression Markers Initiative (PPMI), and because the SAA was done by a laboratory previously shown to have an excellent inter-laboratory agreement with two other reputable groups.²

However, the presented diagnostic accuracy figures were not derived from the standalone SAA results but from the compound accuracy of SAA and dopamine transporter (DAT)-SPECT. Participants meeting all the PPMI clinical inclusion and exclusion criteria for Parkinson's disease were excluded from the SAA sensitivity calculations if they had scans without evidence of dopamine deficiency (SWEDD). To determine the sensitivity for the SAA alone, without the added benefit of a DAT-SPECT screening test, people categorised as SWEDD

should be included in the calculation. Thus, to facilitate comparison of the SAA results from the PPMI study with those previously published of standalone Parkinson's disease biomarkers,³ we have recalculated the sensitivity to include the participants designated as SWEDD.

373 participants with sporadic Parkinson's disease were included in the PPMI study, as well as 54 people categorised as SWEDD and 163 healthy controls. The proportion of SWEDD (14%) is within the typical prevalence range reported in other studies.⁴ In the PPMI study, Table 2 shows that the sensitivity of the SAA for people with sporadic Parkinson's disease (after exclusion of individuals categorised as SWEDD) was 93.3%. Specificity for people designated as SWEDD was 90.7%, which means that 49 participants with SWEDD had a negative result on the SAA, and five had a positive result. After combining the individuals with SWEDD with the people with sporadic Parkinson's disease, 427 participants with sporadic Parkinson's disease were initially included before DAT-SPECT, of whom 353 were classified with a positive result on the SAA. Thus, we calculate a combined sensitivity for the SAA of 82.7%, rather than 93.3%. Specificity, as calculated against the healthy controls group, is unchanged at 96.3%.

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Authors' reply

We are grateful to Blas Couto and colleagues, to Lorraine V Kalia and Tiago A Mestre, and to Thomas Beach and colleagues for their thoughtful responses to our Article.

Couto and colleagues correctly point out that individuals who identify as Latino or Hispanic are over-represented within prodromal participants in our study. The reason for this imbalance is that many individuals with rapid eye movement sleep behaviour disorder in our study came from a single clinical centre in Spain; it does not indicate a meaningful epidemiological association. We agree that better representation of individuals from diverse social and genetic backgrounds would improve the generalisability of PPMI, as well as many other cohort studies.

Kalia and Mestre propose that the subgroup of people with a clinical diagnosis of Parkinson's disease, but with negative α -synuclein seed amplification assay (SAA) results, should be referred to as samples without evidence of synuclein seeding (SWESS), referencing the known term scans without evidence of dopaminergic degeneration (SWEDD). We understand and agree with the points raised by Kalia and Mestre. The challenge of classifying individuals with clinical features of Parkinson's disease (and abnormal dopaminergic imaging), but with negative results on α -synuclein seed amplification assay

(SAA), will provide an opportunity for the field to move from a clinical to a biological definition of Parkinson's disease. We believe this step will be one of the most important contributions of the SAA and will improve understanding of disease biology. We and others are actively engaged in defining a biological classification and staging system for disorders that feature α -synuclein pathology. Kalia and Mestre also comment that a biological definition of disease might be crucial for testing biologically based disease-modifying treatments, but less important for symptomatic treatments in the course of clinical care. We agree with these points, and believe that our work will both provide more effective patient engagement in specific clinical trials and enable targeted treatment in the future, which is essential for a disease that most likely involves multiple pathways.

Beach and colleagues also call attention to the distinction between clinical and biological definitions of Parkinson's disease. They note that the sensitivity for the SAA would be lower if all people with clinical parkinsonism were included in the calculation, rather than only those individuals who had both clinical features of Parkinson's disease and abnormal dopamine transporter imaging. SWEDD participants, who—as noted above—have features of parkinsonism but normal imaging, were excluded. Including SWEDD participants in the sensitivity calculation makes the sensitivity lower than reported in our paper (82.7% vs 93.3%).

The issue raised by Beach and colleagues highlights the low diagnostic accuracy of early Parkinson's disease, an area in which Beach (with Charles Adler) has provided some of the most compelling evidence.² Over time, several participants categorised as SWEDD in PPMI have developed progressive parkinsonism, but many have not, consistent with other longitudinal follow-up studies of such

individuals.³ Thus, we believe that, to best estimate the accuracy of the SAA, it is important to include those people most likely to have Parkinson's disease.

We are heartened that our Article, along with others examining α -synuclein seed amplification across the spectrum of neuronal synuclein diseases, has been a catalyst for a thoughtful reconsideration of these clinical entities,⁴ and the emergence of a framework for biologically directed therapeutic development.⁵

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